

RESEARCH ARTICLE

DISCLAIMER: This article contains examples of offensive and abusive language as a necessary part of illustrating its research findings.

Integrating Sentiment Analysis With Machine Learning for Cyberbullying Detection on Social Media

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ABSTRACT The phenomenon of cyberbullying has emerged as a critical challenge in the digital landscape which poses detrimental effects on individuals and broader societal frameworks. A viable approach to addressing this pervasive issue lies in the accurate identification of cyberbullying within social media as it constitutes a substantial portion of digital interactions. State-of-the-art solutions predominantly rely on pre-trained language models and machine learning algorithms; however, these methods are often associated with substantial computational overheads and the development of advanced cyberbullying detection algorithms remains limited. This paper presents a unique framework that integrates sentiment analysis with machine learning algorithms to enhance the detection of cyberbullying on social media. Sentiment analysis explores the linguistic and emotional characteristics of cyberbullying messages which helps the framework to identify key sentiment indicators that differentiate harmful interactions from benign ones. Furthermore, we provide the most optimal text preprocessing steps which are ordered in a way that improves text quality for cyberbullying detection. These steps ensure high-quality input data for machine learning models which significantly enhances their performance. Additionally, we tackle the challenge of penta-class imbalanced data by incorporating resampling within the framework without causing bias. We employ several machine learning algorithms within the framework, the Extra Tree classifier achieved the best results with an accuracy of 95.38% and F1 score of 0.95. The results demonstrate that the integration of sentiment analysis significantly improves classification accuracy compared to conventional methods for the task of cyberbullying detection.

INDEX TERMS Cyberbullying detection, sentiment analysis, machine learning, resampling.

I. INTRODUCTION

The advancements in technology have enabled online communities to create and respond to user-generated content through social media. While social media provides ease and efficiency, it also poses risks in the form of cyberbullying which is described as the purposeful and recurrent harm

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perpetrated using electronic devices and social media [1]. Such behaviors include spreading rumors [2], posting extremist content toward a group of individuals [3], creating fake profiles to impersonate someone's identity [4], sending threatening messages, and publicly sharing an individual's sensitive information [5]. Figure 1 shows the various forms

of cyberbullying on social media. A 2022 survey by the Pew Research Center informed that 46% of U.S. teens experienced cyberbullying, with teenage girls being the most targeted victims [6]. While the great majority of parents reported bullying at school, 19.2% said it happened through social media [7]. This issue is frequently overlooked by offenders who fail to comprehend the psychological trauma it causes to the victims. Studies have shown that cyberbullying has serious consequences for mental health among teens, such as increased stress, low self-esteem, feelings of hopelessness, and even suicidal thoughts [8]. Academics have used advances in machine learning (ML) to create algorithms capable of detecting abusive content on social media to mitigate the negative consequences of cyberbullying [9]. Criminals utilize these channels to acquire data and information that allows them to carry out their illicit activities online. Furthermore, cyberbullying has long-reaching damage that extends far beyond the online world, affecting the victim's daily life and interpersonal connections. It is in this context that early detection and intervention become crucial, particularly in the arena of English tweets. English is the most common language in online communications which emphasizes the necessity for focused cyberbullying identification systems in this linguistic domain [10]. Doing so addresses a substantial portion of online interactions, thus reducing such incidents by many folds. Early detection of cyberbullying enables parents, educators, and support networks to take interventive measures and provide victims with the emotional and psychological assistance they require [11]. It also prevents online abuse from escalating to an uglier kind. In light of these considerations, incorporating advanced technologies, such as sentiment analysis and supervised learning, in cyberbullying detection is more than just an academic pursuit; it is an essential step toward creating a safe and compassionate digital landscape where individuals can thrive.

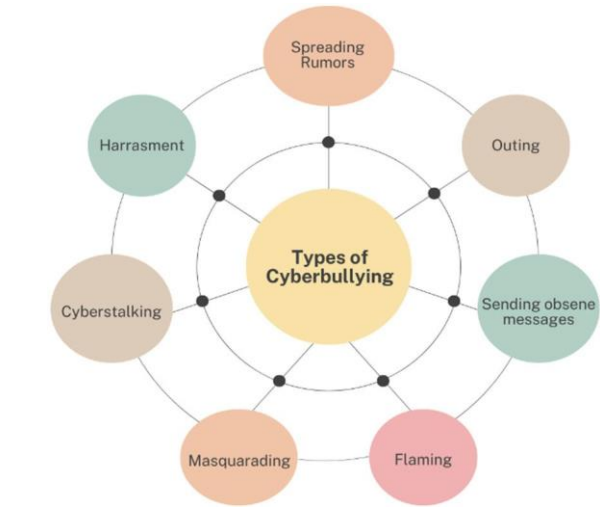


FIGURE 1. Forms of cyberbullying.

Sentiment analysis (SA) is an important tool for the task of cyberbullying detection as it uses natural language processing (NLP) and text analytics to extract subjective material from textual data [12]. It analyzes the tone, emotion, and purpose of online interactions which allows for the discovery of cyberbullying aspects inside social media platforms. Utilizing SA for cyberbullying provides a strong foundation for early detection and intervention because it provides the ability to analyze the emotions in the data. As the frequency of cyberbullying rises, utilizing such new solutions becomes critical to creating safer and more supportive online settings. Researchers have been focusing on developing strategies to identify and mitigate cyberbullying on digital platforms. Recently, cyberbullying detection systems have been getting a lot of attention due to their potential to effectively identify online harassment.

Cyberbullying detection systems are designed to extract specific features from social media messages and employ classification algorithms to detect cyberbullying. These features may include user behavior, message content, emotional tone, or social network dynamics. SA plays a pivotal role in this process by identifying offensive language, harsh remarks, or negative interactions that may signify bullying behavior. Manual monitoring and intervention are impractical because of the immense volume of user-generated content on social media platforms. SA offers a scalable, automated approach to filter through this data and promptly flag potentially harmful posts. Cyberbullying is often characterized by a dominant presence of negative sentiment, particularly in the form of angry, demeaning, or threatening language. SA enables the identification of harmful interactions by detecting these negative emotional patterns even in the absence of explicit bullying language, thereby enhancing the effectiveness of cyberbullying detection systems [13]. For instance, cyberbullying can be identified by patterns of recurrently unfavorable interactions or by minor changes in mood over time. Furthermore, by capturing the tone and context of a discussion, SA can improve identification and provide a more detailed picture of bullying behaviors than keyword matching or cursory text analysis alone. Although a lot of research has been conducted in this domain, there is room for improvement in refining the quality of datasets, enhancing fine-grain detection, and increasing the reliability and reproducibility of models [14]. Additionally, most existing models are based on transformers [15], and deep learning techniques [16], both of which are computationally expensive and time-consuming.

The need to tackle the issue of cyberbullying was emphasized in 2020, when UNICEF delivered a warning about the heightened risk of cyberbullying in the COVID-19 epidemic [17]. Therefore, this research offers a way to handle this challenge through SA and ML-based frameworks. This research provides significant contributions, as stated below:

- We propose a robust multi-step framework that seamlessly integrates text preprocessing techniques with sentiment analysis and machine learning algorithms for cyberbullying detection.
- We analyze the best preprocessing steps for the accurate identification of several categories of cyberbullying. We employ methods such as language filtering, emoji correction, URL, special character and stopword

tools for identifying and mitigating harmful online behavior. SA can be used to extract subjective information from text which enables the detection of aggressive or abusive content. ML and DL algorithms have been successfully applied to text data with promising results in various fields such as sports analytics [18], social media bot detection [19], [20], sentiment analysis [21], [22] and fake news detection [23]. Studies demonstrate that despite the character limit on platforms like Twitter, there is still a necessity for robust

TABLE 1. Cyberbullying sentiment analysis key literature.

Cite	Year	Methodology	Purpose	Findings
[29]	2020	The study uses TF-IDF for feature extraction and Support Vector Machine for classification. Dataset includes 1500 comments for training and testing.	To analyze sentiments in social media comments to detect cyberbullying content.	Achieved 93% accuracy, 95% precision, and 97% recall. Designed a system model that can integrate with browsers for comment classification.
[30]	2023	Various ML models were evaluated on two Twitter datasets. Sentiment Analysis and higher n-gram language models were used for feature extraction.	To create and refine a process for detecting cyberbullying in tweets using SA and ML techniques.	CNN classifiers outperformed other models across both datasets. Traditional ML classifiers performed comparatively lower.
[31]	2023	This study employed Conventional ML and Transfer Learning methods for cyberbullying detection.	The research aimed to develop an automatic system to improve accuracy and reduce computational effort.	The fine-tuned DistilBert model achieved an F-measure of 72.42%, outperforming CML approaches and requiring less feature engineering and computational effort.
[32]	2024	Evaluated ML models using the Bag of Words and Chi-Square feature selection across six categories.	To analyze the effectiveness of various machine learning models in classifying sentiment across multiple cyberbullying categories.	SVM and Logistic Regression achieved the highest accuracy (83%), while Naïve Bayes performed the worst at 62%.
[33]	2024	This study proposed the VADsentence Miner, leveraging the Valence Aware Dictionary and Sentiment Reasoner for sentence-level sentiment analysis using ML techniques.	The research aimed to develop an intelligent mechanism to analyze and polarize cyber-trolling-related tweets efficiently.	The VADsentence Miner outperformed TextBlob with 90% accuracy, 82% precision, 74% recall, and 78% F1-score using Random Forest and TF-IDF.

removal, tokenization, and feature extraction through TF-IDF to ensure that the input data is optimized for efficient classification.

- We deal with the challenge of penta-class imbalanced data through resampling. By classifying cyberbullying into nuanced categories and leveraging techniques like SMOTE resampling to address data imbalance, the research enhances detection accuracy and contributes to understanding the broader implications of online toxicity.

The remaining paper is divided into four sections where section II covers the Literature Review, Section III discusses the material and methods deployed, Section IV includes results, Section V contains Discussion and comparison, and finally, Section VI is the Conclusion.

II. LITERATURE REVIEW

The increasing prevalence of cyberbullying on social media has prompted extensive research into newer detection methods. SA and ML-based techniques have emerged as key

cyberbullying detection mechanisms.

Several approaches have been studied in the field of cyberbullying detection to deal with the ubiquitous problem of online harassment. In [24], the authors conducted a literature survey to compare the algorithmic techniques deployed in this domain. Their finding concluded that in the context of DL, Convolutional Neural Networks (CNNs) yield the best accuracy, while SVMs are the most popular choice for ML, specifically in the linguistic domain of Arabic and Latin. Authors of [25] chose Facebook and Twitter to collect data for their binary cyberbullying detection study and modified their geographical settings to exclude data from Middle Eastern nations. They collected a corpus of Arabic and English data and SVM was trained on the English data which obtained an accuracy of 93.4% and an F1 score of 92.7%. Similarly, SVMs were used to perform binary cyberbullying detection on a dataset of English and Dutch posts obtained from social media sites in [26]. The model gave an F1 score of 64% for English and 61% for Dutch texts.

In [27], authors trained a CNN-BiLSTM on a dataset of multiple languages that were obtained from different open sources, and their model achieved an accuracy of 98%. In a similar study [28], authors developed a Character-level Convolutional Neural Network with Shortcuts to detect cyberbullying instances from English tweets and Chinese Weibo content. Their approach targeted character-level details from the text to overcome spelling mistakes and symbolic errors. The model gave good performance in both languages. Several studies have incorporated SA with cyberbullying detection to enhance its performance. Table 1 lists the key literature related to our topic.

Authors of [34] deal with SA based on the Arabic language, they employed ML techniques such as Support Vector Machines (SVM) to detect cyberbullying on social media. One approach utilized a dataset collected via ArabiTools and the Twitter API, with tweets labeled as cyberbullying or non-cyberbullying based on the existence of aggressive language. Comparative experiments evaluated the performance of WEKA and Python tools. WEKA with Light Stemmer achieved the highest accuracy (85.49%) but required 352.51 seconds for model training. Python tools achieved lower accuracy (84.03%) but significantly reduced training time to 142.68 seconds. These findings emphasize the balance between accuracy and computational efficiency in SA, thus; underscoring the need for tailored preprocessing techniques and tool selection to address cyberbullying detection effectively.

In another study [29], authors utilized a dataset of 1,500 comments collected via Python-based web crawling, processed through preprocessing steps, and TF-IDF for feature extraction. Their ML-based system achieved high-performance metrics, including an accuracy of 93%, a precision of 95%, and a recall of 97%. Additionally, a system design was developed where they integrated a browser-based interface for analyzing and classifying user comments in real time which offered a practical tool for cyberbullying identification and prevention.

Authors of [30] focus on developing and refining a methodology combining SA with advanced language models to detect cyberbullying in tweets. Two datasets of varying sizes were employed to evaluate the performance of multiple ML classifiers. CNNs based on higher-order n-gram language method consistently performed better than other classifiers, demonstrating superior accuracy in identifying abusive language. This research underscores the potential of integrating SA with robust ML techniques to deal with the growing challenge of cyberbullying on social media platforms.

This study [35] utilizes the Toxic Comment Classification Challenge dataset from Kaggle. They apply pre-processing techniques such as comment generalization (lowercasing and punctuation removal), tokenization, stopword removal, and lemmatization. Their SVM-based method undergoes

hyperparameter tuning through Randomized Search CV, with evaluation metrics including Micro F1 Score for prediction accuracy and Hamming Loss for misclassification. The integration of Word2Vec and OAA SVM demonstrated effective results, with 100 features extracted during pre-processing achieving a Micro Averaged F1 Score of 83.40% and a Hamming Loss of 15.13%. These results underscore the potential of advanced sentiment analysis techniques for mitigating cyberbullying.

In another research [36], authors introduce a sentiment analysis model aimed at detecting cyberbullying in tweets. It leverages CNN, SVM, and Naïve Bayes as supervised ML classifiers. The model incorporates higher n-grams language representation to enhance feature extraction where CNN outperformed both NB and SVM in multiple evaluation metrics. The findings highlight the potential of CNN in increasing the detection accuracy of cyberbullying tweets, offering significant advancements over previous methodologies in this domain.

Another study [37] focuses on analyzing sentiment in Indonesian Instagram comments to uncover emotional tendencies and classify comments as positive, negative, or neutral. A dataset of 800 Indonesian comments was collected through web scraping and processed using three classification methods: Several ML classifiers were employed where the NB classifier attained the highest accuracy of 91%, followed by K-NN at 85% and Decision Tree at 75%. Evaluation metrics such as the confusion matrix further validated these findings, demonstrating that Naive Bayes is the most effective method for classifying Instagram comment sentiments in this context.

In [38], the authors propose a cyberbullying detection platform aimed at promoting awareness and enabling users to take proactive measures, such as blocking or reporting accounts involved in cyberbullying. They seek to identify harmful content and foster a safer, more positive online environment by leveraging ML techniques and sentiment analysis. Ultimately, the implementation of such a system has the ability to improve user experiences and contribute to a healthier digital community.

III. MATERIALS AND METHODS

This section details the methodology used for cyberbullying detection leveraging SA and machine learning. The proposed framework, as depicted in Figure 2, encompasses three primary stages: Data Processing, Model Development, and Classification. Each stage is designed to ensure the accurate identification of cyberbullying activities in social media data. The proposed framework ensures a systematic approach to detecting cyberbullying by combining robust preprocessing techniques, effective sentiment analysis, and advanced machine learning algorithms. The proposed framework leverages a comprehensive pipeline to identify abusive content and promote a safer online environment.

A. DATASET DESCRIPTION

We utilize a comprehensive cyberbullying classification dataset presented in [39]. This data is an important resource for the development of an effective framework. The dataset includes 47,692 tweets, which are divided into six main classifications based on the nature of the bullying which are shown in Table 2.

Each class is fairly balanced but there is a slight imbalance in a few classes as shown in Figure 3. We will utilize resampling (section III-E) which will guarantee fair representation for optimal machine learning model training and assessment. The dataset provides a framework for investigating the detection, characterization, and comprehension of cyberbullying in a variety of social situations.

Figure 4 presents a box plot that compares the distribution of word lengths across different types of cyberbullying

Other cyberbullying	Cyberbullying is not categorized under age, ethnicity, gender, or religion.
Not cyberbullying	Tweets that do not contain any cyberbullying or offensive content.

categories. The X-axis represents the length of words (number of words) in a text or sentence while the y-axis shows the different classes of data. The colored boxes represent the middle 50% of the data between the 25th percentile and the 75th percentile. The line inside the box is the median (50th percentile) of the word lengths for that category. The horizontal lines (whiskers) extending from the boxes represent the range of the data, excluding outliers. “Age” and “Religion” categories have the largest word lengths and the highest variability which may reflect more complexordetailed discussionswhile“other_cyberbullying”a nd “not_cyberbullying” contain shorter texts which

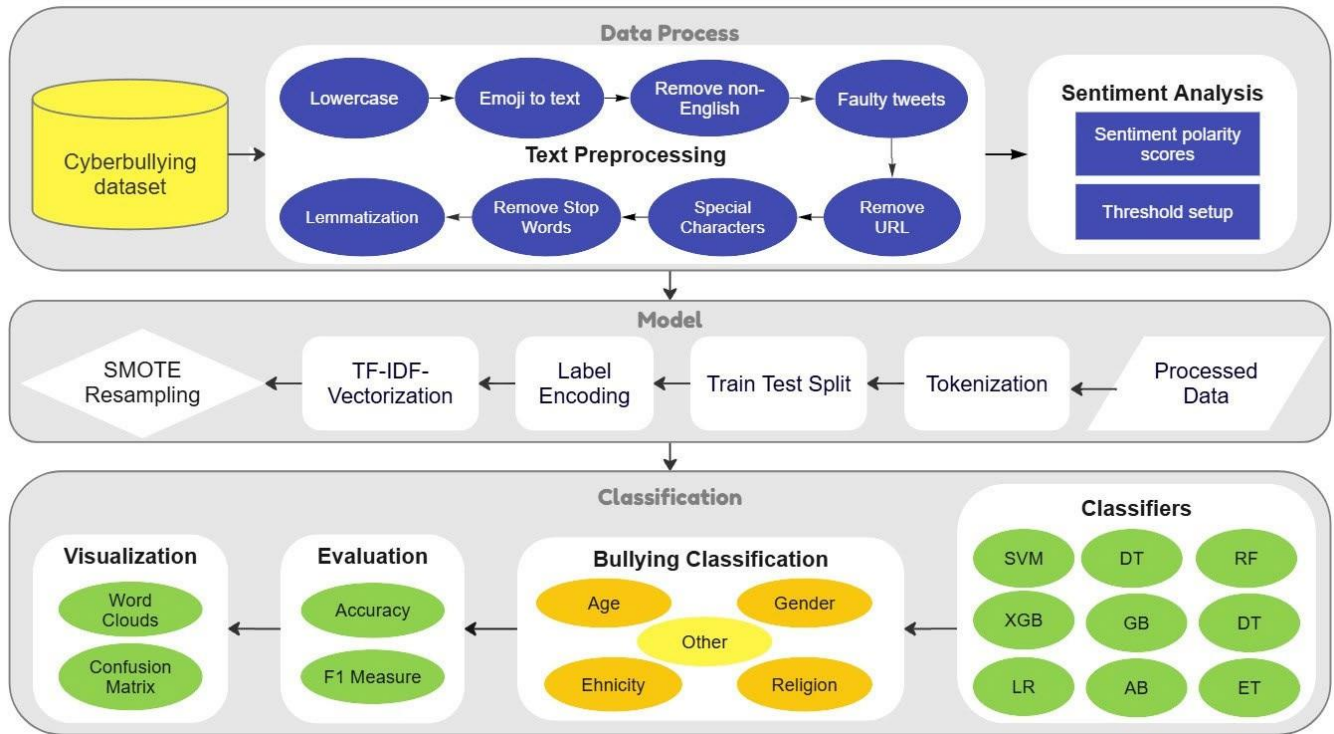


FIGURE 2. Framework for enhanced cyberbullying detection through sentiment analysis and machine learning.

TABLE 2. Cyberbullying categories.

Class	Description
Age	Cyberbullying content is targeted at individuals based on their age.
Ethnicity	Tweets containing discriminatory or bullying content based on ethnicity or race.
Gender	Tweets targeting individuals based on their gender with bullying or offensive content.
Religion	Tweets with content aimed at bullying or discriminating against a person's religion.

indicates simpler or more direct messages. Such visualizations can help

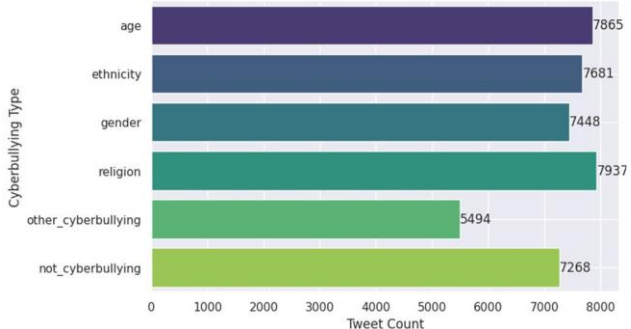


FIGURE 3. Data samples in each class.

us understand the differences in tailoring detection models or improving text-based classification systems.

B. MATHEMATICAL MODELING

Below is a step-wise explanation of the evaluation and working of the proposed model. Table 3 provides the list of mathematical notations used in the model.

Equation 1 defines the objective function (C_{detect}) aims to maximize the accurate detection of cyberbullying presence (Y_{cyber}) or its absence (Y_{non}) in all tweets T_{all} from the dataset (D_{tweets}).

$$C_{detect}(T_{all}) = \max(Y_{cyber}, Y_{non}) \forall T_{all} \in D_{tweets} \quad (1)$$

Equation 2 shows that any empty rows or columns are excluded. Uppercase characters (C_{upper}) are converted to

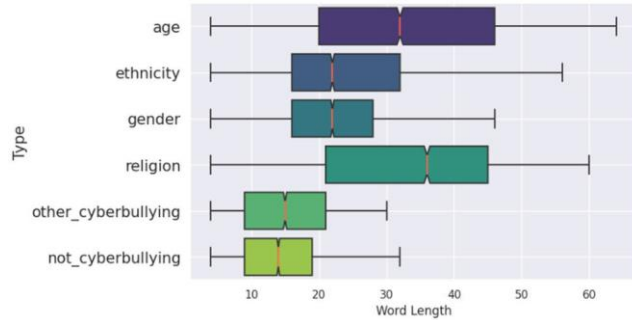


FIGURE 4. Data boxplot.

TABLE 3. Notations and their description for equations.

Notations	Description
C_{detect}	Cyberbullying detection (objective function)
D_{tweets}	The dataset containing all tweets
D_{clean}	Cleaned dataset after removing irrelevant content
$D_{process}$	Preprocessed dataset
T_{all}	All tweets from the dataset
T_{mc}	Misclassified tweets during classification
$S_{polarity}$	Polarity scores from sentiment analysis
E	Emojis
U	URL
C_{upper}	Uppercase
C_{lower}	Lower case
SC	Special characters in tweets
T_{ne}	Non-English tweets
N	Normalization
Y_{cyber}	Presence of cyberbullying in tweets
Y_{non}	Absence of cyberbullying in tweets
SA	Sentiment analysis

lowercase (C_{lower}) to ensure consistency. Furthermore, nonrelevant components such as emojis (E), URLs (U), special characters (SC), and non-English tweets (T_{ne}) are eliminated. Tweets flagged as irrelevant or misclassified (T_{mc}) are excluded from the dataset. $f D_{clean}, D_{process}$

?

Rows/Columns $\neq$ NULL

$$\begin{aligned} n \text{ } ? &= C_{lower} \\ = 6_{i=1} &N_{Cupper} \end{aligned} \quad (2)$$

?

$$T_{mc} = E = U = SC = T_{ne} = 0$$

Equation 3 represents the use of sentiment analysis (SA) on all tweets (T_{all}) to compute polarity scores ($S_{polarity}$). Sentiment analysis ensures that misclassified tweets (T_{mc}) are corrected or removed.

$$SA(T_{all}) \Rightarrow T_{mc} = 0 \quad (3)$$

Tweets with polarity scores between (0,1) are classified as non-cyberbullying (Y_{non}) while tweets with polarity scores between (-1,0) are flagged as cyberbullying (Y_{cyber}) as shown in Eq.4.

$S_{polarity}$

$$\begin{aligned} &= -01,,10 \\ &\quad \text{if } Y_{non} \text{ if} \\ &\quad Y_{cyber} \end{aligned} \quad (4)$$

C. DATA PREPROCESSING

The Twitter-based cyberbullying detection dataset (section III-A) was thoroughly cleaned and preprocessed to get accurate and trustworthy findings. We began the preprocessing operation by looking for empty rows and columns, but there were none found. Following that, a normalization phase was introduced to improve the consistency and convenience of analysis. All tweets were converted to lowercase to eliminate the chance of case variations impacting further text processing. In the following stage, all emojis in the dataset were translated to their corresponding meanings. This decision was inspired by the observation that most internet users prefer using emoticons to establish the tone of their discussions, and deleting them would result in a loss of crucial information. Non-English tweets were excluded due to the emphasis on the English language domain as it helps reduce inefficient characteristics from the dataset. Furthermore, URLs and special characters were eliminated methodically to guarantee that the tweets were

completely refined. Tweets go through a second level of preparation when whitespace tokenization is used to separate the text into its components. Stop words are then removed as they do not provide any meaningful information to the task of cyberbullying detection which helps reduce the dimensionality of the feature space. Finally, the remaining words are lemmatized to unify them with their root forms. This prepares the data for TF-IDF vectorization which assigns weights to words based on their significance within the tweet relative to the entire dataset. These preprocessing

steps streamlined the dataset and prepared it for sentiment analysis.

D. SENTIMENT ANALYSIS

SA focuses on extracting and understanding the emotional tone or sentiment conveyed in the text. In the context of cyberbullying detection, it helps determine whether the tweets or posts carry abusive or offensive language that might be indicative of bullying behavior. Sentiment polarity measures the emotional tone of a given text on a spectrum ranging from negative to positive. For example, positive sentiment indicates supportive and encouraging language while negative sentiment suggests abusive or offensive language. We calculate the polarity score is calculated for each text entry. This score is typically a numeric value where a threshold is set to classify the text into predefined categories like positive, neutral, or negative. Threshold values are critical for determining whether a comment falls into the abusive or cyberbullying category. For instance, if a sentiment score below predefined threshold is observed, the system may flag the text as potentially harmful or offensive.

For our framework, we set the threshold to ‘0’ and remove any tweets that have a polarity score of zero or above, as such tweets either carry neutral, positive sentiment or lack emotionally charged language which makes them less likely to involve harmful or abusive content. This is important because a low threshold may degrade label accuracy, increasing the

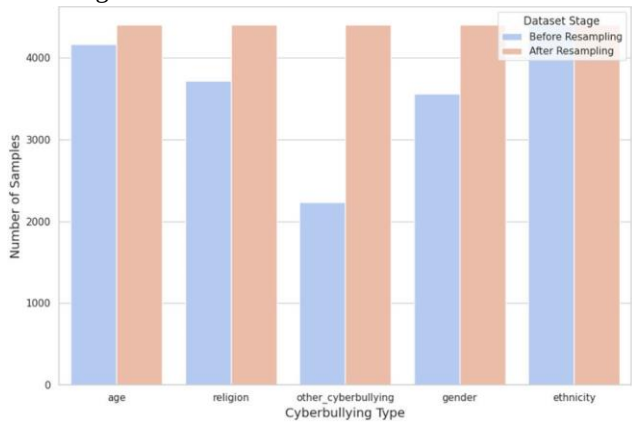


FIGURE 5. Class distribution before and after SMOTE resampling.

probability of false positives while a high threshold may result in more false negatives [40]. Filtering out such data allows our framework to focus on text that is more likely to contribute meaningful patterns for cyberbullying detection. Thus, retaining tweets with negative polarity minimizes the risk of flagging benign tweets as abusive. This improves the precision of the detection system. The use of SA helps to reduce false positives by distinguishing between harmless

negativity (e.g., constructive criticism) and genuinely abusive content.

SA captures the sentiment intensity and emotional valency behind a text giving output as an array of four scores: negative, positive, neutral, and compound. The negative, positive, and neutral scores numerate the strength of negativity, positivity, and neutrality in the text, whereas the combined score defines the overall sentiment of the text.

E. RESAMPLING

Imbalanced datasets lead to biased machine learning models where the classifier tends to favor the majority class [41]. SMOTE (Synthetic Minority Over-sampling Technique) creates new samples for minority classes to balance the dataset [42]. Figure 5 illustrates a bar chart comparing the class distributions before and after SMOTE resampling. The X-axis represents the various categories of cyberbullying types while the y-axis represents the number of samples for each class. It confirms that all classes now have equal sample sizes. The blue bars indicate an imbalance in the dataset before resampling, with a significantly lower number of samples for the “other cyberbullying” category compared to others while the orange bars show the distribution after applying SMOTE oversampling which balances the data by increasing the sample size in underrepresented categories. This balance improves the model’s ability to generalize and predict underrepresented classes effectively. While balancing the dataset helps improve performance for minority classes, it’s essential to monitor for overfitting or synthetic data noise in the model. Therefore, to avoid the issue of overfitting we only apply resampling to the training set.

F. ML CLASSIFICATION

The ML classification phase forms the backbone of the proposed framework for cyberbullying detection by utilizing multiple ML algorithms to identify offensive behavior across penta-class cyberbullying. The classification model leverages the preprocessed data which is enriched with SA to train and evaluate classifiers for optimal performance. To address imbalanced datasets, both unprocessed and SMOTE-oversampled data were considered, enabling a comparative analysis of classifier performance in imbalanced and balanced scenarios. Initially, the data is portioned into training and testing sets with a ratio of 70% to 30% ensuring an appropriate split for robust model evaluation. Label encoding is then applied to convert categorical data into numerical labels. Nine ML classifiers were evaluated in this study: Logistic Regression (LR), Random Forest (RF), XGBoost (XGB), Decision Tree (DT), Naive Bayes (NB), Support Vector Machine (SVM), Gradient Boosting (GB), and Extra Tree (ET) and AdaBoost. Each classifier was optimized using grid search for hyperparameter tuning, ensuring enhanced accuracy and generalization. The evaluation

metrics include accuracy and F1-score, emphasizing balanced performance across different cyberbullying categories.

IV. RESULTS

The results of integrating SA with ML classifiers for cyberbullying identification on social media demonstrate varying levels of performance considering the ML classifier used and the use of resampling. Table 4 shows the results of all classifiers with and without SMOTE oversampling. Without resampling, the combination of SA and SVM classifier achieved the highest accuracy of 95.35% and an F1 score of 0.95 which is closely followed by the Extra Tree (ET) classifier with an accuracy of 95.33% and an F1 score of 0.95. The LR model also exhibited competitive performance, reaching an accuracy of 95.23% and an F1 score of 0.95. Other classifiers such as RF and GB, yielded accuracy values of 93.14% and 94.73% respectively, showcasing their reliability in detecting cyberbullying.

After applying SMOTE to address the class imbalance, the ET Classifier outperformed all other models by achieving an accuracy of 95.38% and an F1 score of 0.95 which indicates its robustness even in a balanced dataset. The SVM classifier demonstrated slightly reduced performance with an accuracy of 95.12%, while Logistic Regression achieved 94.82% accuracy, maintaining their respective F1 scores of 0.95. The Naive Bayes (NB) classifier showed significant improvement after resampling, increasing its accuracy from 87.39% to 90.30% and its F1 score from 0.86 to 0.90. Similarly, classifiers such as RF and XGBoost (XGB) exhibited improved performance by achieving F1 scores of 0.94 post-SMOTE.

The results for the Extra Tree (ET) classifier in Table 5 highlight the performance of the model across different categories of cyberbullying. The model demonstrates near-perfect performance for detecting cyberbullying related to ‘‘age’’,

TABLE 4. Results of framework with ML classifiers.

	No Resampling		SMOTE	
	Accuracy	F1	Accuracy	F1
SA + LR	95.23%	0.95	94.82%	0.95
SA + RF	93.14%	0.93	94.21%	0.94
SA + XGB	94.73%	0.95	94.07%	0.94
SA + DT	93.51%	0.93	93.29%	0.94
SA + NB	87.39%	0.86	90.30%	0.90
SA + SVM	95.35%	0.95	95.12%	0.95
SA + Extra Tree	95.33%	0.95	95.38%	0.95
SA + GB	94.73%	0.95	94.01%	0.94
SA + AdaBoost	91.61%	0.92	90.46%	0.91

‘‘ethnicity’’ and ‘‘religion’’. High precision indicates minimal false positives, and high recall suggests effective detection of true positives. Performance is slightly lower for gender-based cyberbullying while the ‘‘other_cyberbullying’’ has the lowest performance metrics where the model struggles to detect or classify non-standard or less-defined forms of cyberbullying.

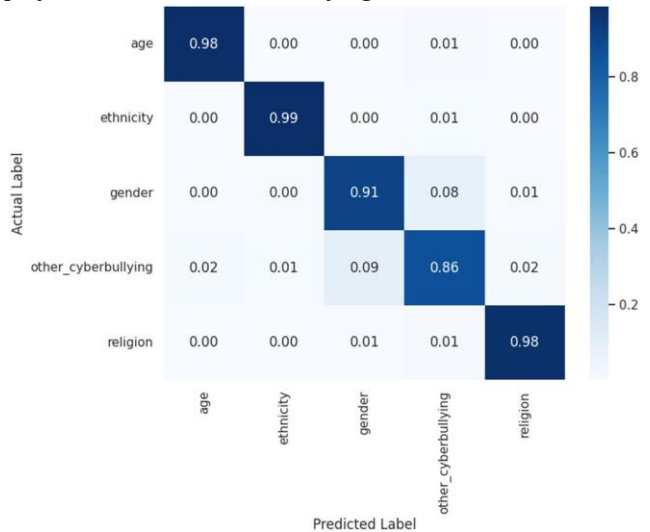
TABLE 5. Category-wise results for the ET classifier.

Cyberbully category	Precision	Recall	F1-score
age	0.99	0.98	0.98
ethnicity	0.99	0.99	0.99
gender	0.92	0.91	0.92
religion	0.98	0.97	0.98
other-cyberbullying	0.81	0.85	0.83

Overall, the findings demonstrate that the ET Classifier when combined with SA consistently provides the most accurate results for cyberbullying detection across both imbalanced and balanced datasets. While SMOTE generally enhanced performance for minority class detection, its effect was marginal for models like SVM and Extra Tree, which already performed exceptionally well on the imbalanced dataset. Figure 6 shows the confusion matrix for the best-performing ET classifier. These findings emphasize the importance of leveraging robust preprocessing techniques and resampling methods in combination with SA and ML algorithms to effectively detect cyberbullying on social media platforms.

V. DISCUSSION AND COMPARISON

This research aims to explore the effectiveness of combining SA and ML techniques to detect cyberbullying in text data on social media. The results obtained from the experiments demonstrate the potential and limitations of current approaches and provide insights into the challenges of developing robust systems for detecting abusive content. SA played a critical role in identifying the emotional tone of



text which enabled the detection of negative and aggressive sentiments commonly associated with cyberbullying. The preprocessing steps ensured that the extracted features were relevant and representative of the data. We utilized sentiment polarity as a filtering mechanism which reduced noise and allowed the models to focus on text more likely to contain harmful content.

Errors in the scoring system could lead to the exclusion of tweets with subtle negative tones. Therefore, it is important to consider this drawback when dealing with real-world cyberbullying detection.

We tested several ML classifiers as they provide faster training times and better interpretability. These models could be useful in scenarios with limited computational resources or for applications requiring real-time responses. The integration of advanced techniques like SMOTE resampling and hyperparameter tuning further enhanced model performance. Existing approaches implemented on the cyberbullying detection dataset used in this study are reviewed and presented in Table 6. It can be seen that a

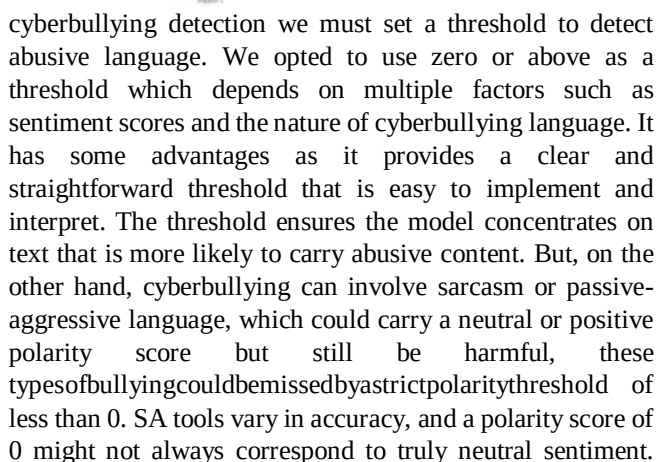


FIGURE 7. Word clouds of all categories of cyberbullying.

Cite	Technique	Accuracy	F1
39	SOSNet + SBERT	92.7%	0.93
[43]	CNN, RNN, Congruent attention, Dynamic routing	93.47%	0.93
44	LightGBM, XGBoost, RF, LR	85.5%	0.85

45	NB, DT, and SVMs	-	0.93
Ours	SA + ML + SMOTE	95.38%	0.95

have been used for cyberbullying detection including ML, DL, and transformer-based approaches. Previous research has used a wide range of strategies and ways to combat the prevalent issue of online harassment have been developed in the field of cyberbullying detection and they achieved significant results. Our framework provides better performance in comparison with an accuracy of 95.48% and F1 score of 0.95.

The findings from this research have significant implications for the development of cyberbullying detection methods. Social media platforms can implement proactive measures to identify and address harmful behavior by integrating sentiment analysis with ML. The inclusion of a threshold mechanism for sentiment polarity filtering ensures that the system is effective in handling large volumes of data while maintaining accuracy. However, practical deployment would require the model to operate in real time and adapt to dynamic language patterns. Additionally, incorporating multilingual capabilities would enhance the system's applicability across diverse user bases. Therefore, further comprehensive datasets are needed for the task of cyberbullying detection.

A. VISUALIZATIONS

Word clouds were generated to better understand the distinctive linguistic patterns within each class. These visual representations provide in-depth details about the most common terms associated with each class. The larger a word appears in the cloud, the more frequently it occurs in that class. By examining these word clouds, dominant keywords that define cyberbullying within each class can be identified. Thus, this analysis helps explain the model's predictions and delivers vital intuitions into the nature of online harassment across multiple dimensions, contributing to a more thorough understanding of the phenomena.

Figure 7 shows the word clouds of all categories of the data. The category "not cyberbullying" represents words commonly used in conversations not associated with cyberbullying. Terms like "school" and "class" without any negative sentiment suggest topics related to general discussion or non-threatening content. This distinction is essential in a cyberbullying detection model to avoid false positives by identifying non-cyberbullying conversations accurately. The word cloud for gender-based cyberbullying includes terms such as "gay," "joke," "rape," "female," and "bitch," which are often associated with derogatory comments, sexual harassment, and rude language. The presence of these words highlights the prevalence of gender-based harassment online. The sentiment analysis-

based model benefits from detecting negative sentiment and aggression associated with these gendered terms to improve classification accuracy for this category. Terms like "Muslim," "terrorist," "Islam," "Christian," and "support" dominate the category "religion" which suggests discussions or attacks related to religious identity and beliefs. Cyberbullying detection models account for these words as they often carry strong negative sentiments or are used in religious-based cyberbullying. The word cloud 'ethnicity' shows highly offensive and racial slurs, including "nigger," "ass," and "black." Ethnicity-based cyberbullying contains highly aggressive language and our model is sensitive to these terms while distinguishing between offensive and neutral use. The category 'age' includes words like "middle school," "high school," and "bully," reflecting bullying related to age or school-level conflicts. Age-based cyberbullying often involves younger populations and is critical in detecting harm among teenagers. The detection model can identify words associated with age to better protect vulnerable age groups from bullying. Words like "idiot," and "feminazi" suggest other types of bullying that do not fit neatly into the previous categories but still contain elements of insult, mockery, or aggression. It may include general cyberbullying or language used to demean individuals without targeting a specific demographic. Our model analyzes the tone and context of such words to accurately classify them as cyberbullying. The use of sentiment analysis in detecting cyberbullying is a key aspect because negative sentiment is often present in cyberbullying content. Words with strong negative connotations, insults, or offensive language can help identify cyberbullying instances across these different categories. This analysis shows the importance of customizing sentiment analysis to recognize context-specific language, especially offensive words that might be ignored in neutral sentiment analysis but are crucial in cyberbullying detection.

VI. CONCLUSION

Social media is an important part of daily communication; however, this connectivity has also facilitated a concerning rise in cyberbullying. To address this issue, we employed an SA-enhanced ML-based framework to detect cyberbullying. The integration of SA proved instrumental in capturing the emotional tone of text which allows the model to focus on negative and aggressive language commonly associated with cyberbullying. Furthermore, preprocessing steps were implemented to improve the quality of the text which improved the accuracy of the framework. The preprocessing pipeline included converting text to lowercase, removing emojis, non-English text, special characters, URLs, and stop words, as well as performing lemmatization. These steps ensured that the textual data was cleaned and standardized which allowed the framework to focus on meaningful features rather than

noise. Furthermore, SMOTE resampling was employed to deal with the issue of imbalanced class in the dataset which is a common challenge in cyberbullying detection tasks. This step significantly reduced the risk of bias toward majority classes which ensured a more equitable classification process.

The ET classifier outperformed most in terms of accuracy and F1 when compared to other ML classifiers. The Extra Tree Classifier demonstrated remarkable performance in identifying harmful content, achieving an accuracy of 95.38% and an F1 score of 0.95. Despite the promising results, this research acknowledges the inherent complexities of cyberbullying detection. Sarcasm and cultural nuances remain challenging for ML models even with SA integration.

As the digital landscape evolves, so must our techniques for combating cyberbullying; therefore, we present future directions to enhance cyberbullying detection. Cyberbullying often relies on context and patterns of interaction rather than isolated textual content. Incorporating user behavior and conversational context into the detection process could enhance accuracy and reduce false positives. Additionally, future research could benefit from combining text-based SA with other modalities, such as images, videos, and metadata to enhance the performance.

CONTRIBUTIONS

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